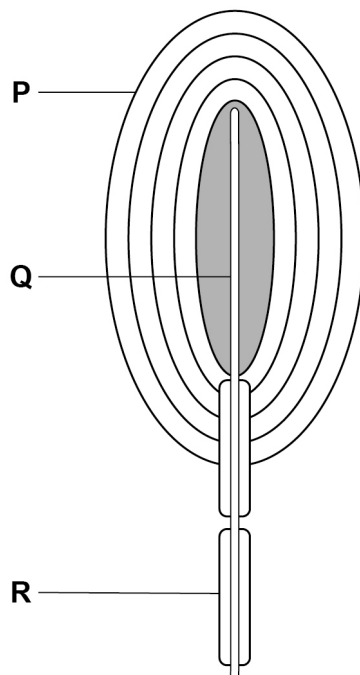


0	2
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Figure 4 shows a diagram of a Pacinian corpuscle.

Figure 4



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Name the structures labelled **P**, **Q** and **R** shown in **Figure 4**.

[2 marks]

P _____

Q _____

R _____

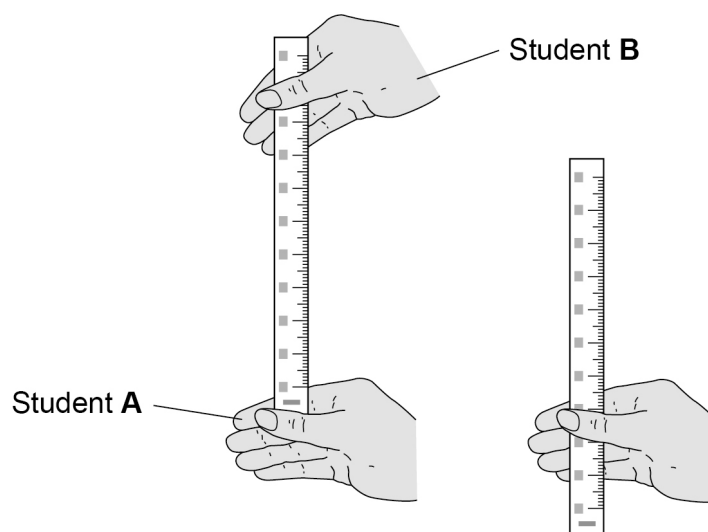
Question 2 continues on the next page

Turn over ►

Two students (**A** and **B**) investigated reaction time in response to touch.

- Student **A** sat with her eyes shut and her forearm resting on a worktop so that her hand was over the edge.
- Student **B** held a ruler vertically between student **A**'s thumb and first finger, with the ruler at 0 mm lightly touching student **A**'s first finger.
- Student **B** released the ruler.
- As soon as student **A** felt the ruler fall, she closed her thumb and first finger to catch the ruler as shown in **Figure 5**.
- Student **B** measured the distance the ruler had fallen to the nearest mm

Figure 5



The test was repeated three more times using the same hand to catch the ruler. **Table 3** shows student **A**'s results.

Table 3

Trial	Distance the ruler has fallen / mm
1	79
2	97
3	10
4	94

The student was able to convert these distances into reaction times using **Table 4**.

Table 4

Distance the ruler fell / mm	Reaction time / ms
10	45
20	64
30	78
40	90
50	101
60	111
70	120
80	128
90	136

0 2 . 2 Calculate the percentage uncertainty in the measurement of **Trial 1** in **Table 3**.

Put a tick (✓) in the correct box below.

[1 mark]

0.633%

☐

1.27%

☐

2.53%

☐

12.6%

☐

0 2 . 3 In this investigation, it is not possible for a student to react in less than 45 ms

Suggest **one** explanation for the value recorded in **Trial 3** in **Table 3**.

[1 mark]

0 2 . 4 Student **A** estimated that the length of the nerve pathway involved was 175 cm

Use **Table 3** and **Table 4** to calculate the mean speed of nerve impulse transmission.

Do **not** use the value for **Trial 3** in your calculation.

[2 marks]

Answer _____ m s⁻¹

Question 2 continues on the next page

Turn over ►

0	2	.	5
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In response to touch, nerve impulses can be transmitted at speeds of 76.2 m s^{-1}

Suggest **three** reasons why, in this investigation, the estimated speed of student **A**'s impulse transmission was less than 76.2 m s^{-1}

[3 marks]

1 _____

2 _____

3 _____
